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Assistance apparatus, vehicle, computer readable storage medium, and assistance method

Abstract

An assistance apparatus includes a reception unit configured to receive information indicating a travelling hazardous point which is transmitted from an outside of a vehicle, a calculation unit configured to calculate a distance and an orientation from the vehicle to the point based on the point and a current position of the vehicle, a detection unit configured to detect the point inside a detection range set within a range in which a distance from the vehicle is a first distance or more and a second distance or less, and an assistance control unit configured to perform travelling assistance when the point is detected inside the detection range, in which the first distance is a travel distance required for a speed of the vehicle to be slowed down to a predetermined speed or less by the time the vehicle reaches the point from the current position of the vehicle.

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Background/Summary

(1) The contents of the following Japanese patent application(s) are incorporated herein by reference: NO. 2022-052491 filed on Mar. 28, 2022.

BACKGROUND

1. Technical Field

(2) The present invention relates to an assistance apparatus, a vehicle, a computer readable storage medium, and an assistance method.

2. Related Art

(3) In recent years, efforts have been intensified to provide an access to a sustainable transportation system with consideration also given to vulnerable people among traffic participants. To this end, research and development has been focused on to further improve traffic safety and convenience through research and development regarding a preventive safety technology. Patent Documents 1 and 2 describe a technique related to a range for detecting an object or another vehicle.

LIST OF CITED REFERENCES

(4) Patent Document 1: Japanese Patent Application Publication No. 2015-45622 Patent Document 2: Japanese Patent Application Publication No. 2011-253241

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 schematically illustrates a usage scene of an assistance apparatus **40** according to an embodiment.

(2) FIG. 2 illustrates a state in which a vehicle **20** travels in a direction of travel.

(3) FIG. 3 illustrates a system configuration of the vehicle **20**.

(4) FIG. 4 illustrates a state in which a detection range **400** is changed according to a speed.

(5) FIG. 5 illustrates a detection range **500** which is to be set when the vehicle **20** is travelling on a curved road.

(6) FIG. 6 illustrates a data structure of detection range data which is referred to when a range change unit **220** sets a detection range.

(7) FIG. 7 illustrates an example of a computer **2000**.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(8) Hereinafter, the present invention will be described by way of embodiments of the invention, but the following embodiments are not intended to limit the invention according to the claims. In addition, not all of the combinations of features described in the embodiments are essential for a solving means of the invention.

(9) FIG. 1 schematically illustrates a usage scene of an assistance apparatus **40** according to an embodiment. The vehicle **20** includes an assistance apparatus **40**. The vehicle **20a** includes an assistance apparatus **40a**. The vehicle **20b** includes an assistance apparatus **40b**. The assistance apparatus **40**, the assistance apparatus **40a**, and an assistance apparatus **40b** have a communication function, and are configured to provide vehicle-to-vehicle communication of the vehicle **20**, the vehicle **20a**, and the vehicle **20b**. The assistance apparatus **40**, the assistance apparatus **40a**, and the assistance apparatus **40b** have a travelling assistance function of performing control related to travelling assistance of the respective vehicles.

(10) The vehicle **20a** is a vehicle travelling ahead on a travelling path of the vehicle **20**. The vehicle **20b** is a vehicle travelling on a path different from the travelling path of the vehicle **20**. The assistance apparatus **40**, the assistance apparatus **40a**, and the assistance apparatus **40b** can inform another vehicle of slip points which are points at which the respective vehicles have slipped.

(11) In FIG. 1, when it is sensed that the vehicle **20a** has slipped because of a puddle or the like during travelling, the assistance apparatus **40a** transmits, to surrounding vehicles, slip point information indicating a slip point **110** that is a point at which the vehicle **20a** has slipped. When it

is sensed that the vehicle **20b** has slipped because of a puddle or the like during travelling, the assistance apparatus **40b** transmits, to surrounding vehicles, slip point information indicating a slip point **120** that is a point at which the vehicle **20b** has slipped. The slip point information may include latitude and longitude information of the slip point.

(12) It is assumed that the vehicle **20** is travelling on a straight road. When the slip point information is received, the assistance apparatus **40** sets a detection range **100** for detecting the slip point. The detection range **100** is a range located ahead of the vehicle **20** along the direction of travel of the vehicle **20** relative to a current position **P0** of the vehicle **20**. Specifically, the detection range **100** is a range obtained by excluding a fan-shaped range with a central angle $\theta 1$ which is surrounded by a straight line **P0P3**, an arc with a radius **L2**, and a straight line **P0P4** from a fan-shaped range with the central angle $\theta 1$ which is surrounded by a straight line **P0P1**, an arc with a radius **L1**, and a straight line **P0P2**. Herein, **P3** is on the straight line **P0P1**, and **P4** is on the straight line **P0P2**. The detection range **100** is a range bilaterally symmetrical with respect to the direction of travel of the vehicle **20**.

(13) When the vehicle **20** has slowed at a predetermined deceleration rate, **L2** is a distance for the vehicle **20** to travel until the speed of the vehicle **20** is slowed down to a predetermined speed. In this manner, **L2** is set according to the vehicle speed of the vehicle **20**. Similarly as in **L2**, **L1** is set according to the vehicle speed of the vehicle **20**. **L1** is set to be longer than **L2**.

(14) When the slip point **110** is inside the detection range **100**, the assistance apparatus **40** performs the travelling assistance. On the other hand, when the slip point **110** is not inside the detection range **100**, the assistance apparatus **40** does not perform the travelling assistance. As illustrated in FIG. 1, since the slip point **110** and the slip point **120** do not exist inside the detection range **100**, the assistance apparatus **40** does not perform the travelling assistance.

(15) Note that in a case where the travelling assistance is performed, even when new slip point information is received from another vehicle, a configuration may be adopted where the assistance apparatus **40** continues the travelling assistance until the vehicle **20** passes through the slip point **110** and does not perform travelling assistance based on information indicating a new travelling hazardous point.

(16) Note that when the slip point is sensed, the assistance apparatus **40** may transmit slip point information indicating the slip point that is sensed to the outside of the vehicle **20**. When the slip point is sensed, a configuration may be adopted where the assistance apparatus **40** does not perform the travelling assistance.

(17) FIG. 2 illustrates a state in which the vehicle **20** travels in the direction of travel. As the vehicle **20** further travels, when the slip point **110** enters the detection range **100**, the assistance apparatus **40** performs the travelling assistance. For example, the assistance apparatus **40** outputs an alert to a driver of the vehicle **20** to beware of slipping.

(18) As a result, it is possible to perform notification when the vehicle speed of the vehicle **20** can be slowed down to the predetermined speed by the time the vehicle **20** reaches the slip point **110**. Furthermore, it is possible to suppress output of an unnecessary alert related to the slip point **120**.

(19) FIG. 3 illustrates a system configuration of the vehicle **20**. The vehicle **20** includes a sensor **29**, the assistance apparatus **40**, a communication apparatus **250**, and an alert apparatus **270**.

(20) The sensor **29** includes a yaw rate sensor **24**, a GNSS receiver **25**, a vehicle speed sensor **26**, and the like. The vehicle speed sensor **26** is configured to detect a vehicle speed of the vehicle **20**. The yaw rate sensor **24** is configured to detect a yaw rate of the vehicle **20**. The GNSS receiver **25** is configured to acquire information of a current position of the vehicle **20**. The alert apparatus **270** is equipment including, for example, a human machine interface (HMI) function. The communication apparatus **250** is responsible for communication with the outside of the vehicle **20**. The communication apparatus **250** is configured to perform vehicle-to-vehicle communication by direct communication such as, for example, PC5. The communication apparatus **250** includes a reception unit **251** and a transmission unit **252**.

(21) The assistance apparatus **40** includes a processing unit **200** and a storage unit **290**. The processing unit **200** is realized by an arithmetic processing apparatus including a processor, for example. The storage unit **290** is realized by including a nonvolatile storage medium. The processing unit **200** performs processing by using information stored in the storage unit **290**. The processing unit **200** may be realized by an electronic control unit (ECU) including a microcomputer equipped with a CPU, a ROM, a RAM, an I/O, a bus, and the like.

(22) The processing unit **200** includes a calculation unit **210**, a range change unit **220**, a detection unit **230**, and an assistance control unit **240**.

(23) The reception unit **251** is configured to receive information indicating a travelling hazardous point which is transmitted from the outside of the vehicle **20**. The slip point is an example of the travelling hazardous point. The travelling hazardous point may be any point with a risk for the travelling of the vehicle in addition to the slip point.

(24) The calculation unit **210** is configured to calculate a distance and an orientation from the vehicle **20** to the travelling hazardous point based on the travelling hazardous point and the current position of the vehicle **20**. The detection unit **230** is configured to detect the travelling hazardous point inside a detection range set within a range in which the distance from the vehicle **20** is a first distance or more and a second distance or less. The first distance is a travel distance required for a speed of the vehicle **20** to be slowed down to a predetermined speed or less by the time the vehicle **20** reaches the travelling hazardous point from the current position of the vehicle **20**. The assistance control unit **240** is configured to perform the travelling assistance when the travelling hazardous point is detected inside the detection range. For example, the assistance control unit **240** informs that the travelling hazardous point exists through the HMI function included in the alert apparatus **270**. Note that the assistance control unit **240** may assist driving of the vehicle **20** itself. The assistance control unit **240** may assist control of the travelling speed of the vehicle **20**. The assistance control unit **240** may assist steering of the vehicle **20**.

(25) The range change unit **220** may set a spread angle of the detection range from the current position of the vehicle **20** to be larger and set the second distance to be longer as the travelling speed of the vehicle **20** is faster. The range change unit **220** may change a width of the detection range depending on whether the travel path of the vehicle **20** is a straight road or a curved road. For example, in a case where the travel path of the vehicle **20** is a curved road, the range change unit **220** may set the detection range to be wider than a case where the travel path of the vehicle **20** is a straight road. In a case where the travel path of the vehicle **20** is a curved road, the range change unit **220** may set the spread angle of the detection range from the current position of the vehicle **20** to be larger than a case where the travel path of the vehicle **20** is a straight road.

(26) When it is sensed that a point at which the vehicle **20** has travelled is hazardous for travelling, the transmission unit **252** is configured to transmit, to the outside of the vehicle **20**, information indicating the point that is sensed to be hazardous for travelling as information indicating the travelling hazardous point. For example, when slipping is detected by an ABS function, the assistance apparatus **40** may sense that the point at which the vehicle **20** has travelled is hazardous for travelling. When it is sensed that the point at which the vehicle **20** has travelled is hazardous for travelling, the assistance control unit **240** may suppress the travelling assistance.

(27) When performing the travelling assistance is performed, even when the reception unit **251** receives information indicating a new travelling hazardous point, the assistance control unit **240** may continue the travelling assistance until the vehicle **20** passes through the travelling hazardous point and may suppress the travelling assistance based on the information indicating the new travelling hazardous point.

(28) FIG. 4 illustrates a state in which a detection range **400** is changed according to a speed. The detection range **400** is a detection range in a case where the vehicle **20** is travelling on a straight road and the vehicle speed of the vehicle **20** is faster than a case illustrated in FIG. 1. Note that the range change unit **220** can determine whether the vehicle **20** is travelling on a straight road based

on the speed detected by the vehicle speed sensor **26** and the yaw rate detected by the yaw rate sensor **24**.

(29) Specifically, the detection range **400** is a range obtained by excluding a fan-shaped range with a central angle $\theta 1'$ which is surrounded by a straight line $P0P3'$, an arc with a radius $L2'$, and a straight line $P0P4'$ from a fan-shaped range with the central angle $\theta 1'$ which is surrounded by a straight line $P0P1'$, an arc with a radius $L1'$, and a straight line $P0P2'$. Herein, $P3'$ is on a straight line $P0P1'$, and $P4'$ is on a straight line $P0P2'$. The detection range **400** is a range bilaterally symmetrical with respect to the direction of travel of the vehicle **20**.

(30) The central angle $\theta 1'$ is an angle formed by straight lines defining the detection range **400** that are left and right straight lines with respect to the direction of travel of the vehicle **20**. That is, the central angle $\theta 1'$ represents the spread angle of the detection range **400**. $L1'$ is a distance from the current position **P0** of the vehicle **20** to a farthest position of the detection range **400** from the vehicle **20**. The range change unit **220** sets $L1'$ to be longer than $L1$ and sets $L2'$ to be longer than $L2$. In addition, the range change unit **220** sets $\theta 1'$ to be smaller than $\theta 1$. As a result, as the speed of the vehicle **20** is faster, a spread of the detection range in a direction orthogonal to the travelling direction of the vehicle **20** is set to be narrower. In this manner, the range change unit **220** changes the spread angle of the detection range according to the vehicle speed of the vehicle **20**.

(31) The detection unit **230** determines whether the slip point is included in the detection range **400** based on the distance and the orientation to the slip point **110** from the vehicle **20** which are calculated by the calculation unit **210** and also based on the current position of the vehicle **20**. When the slip point **110** is included inside the detection range **400**, the assistance control unit **240** performs the travelling assistance.

(32) In normal circumstances, as the speed of the vehicle **20** is faster, it is less likely that the travelling direction of the vehicle **20** suddenly changes in a crosswise direction. Thus, as the speed of the vehicle **20** is faster, the detection range in the direction orthogonal to the travelling direction of the vehicle **20** is set to be narrower, so that it is possible to suppress unnecessary output of an alert.

(33) Similarly as in $L2$, when the vehicle **20** is slowed at a predetermined deceleration rate, the range change unit **220** changes $L2'$ to a distance for the vehicle **20** to travel until the speed of the vehicle **20** is slowed down to a predetermined speed. In this manner, $L2'$ is set according to the vehicle speed of the vehicle **20**. As a result, an alert can be output only when the vehicle **20** exists within a range in which the speed of the vehicle **20** can be slowed down to the predetermined speed.

(34) FIG. 5 illustrates a detection range **500** which is to be set when the vehicle **20** is travelling on a curved road. Herein, it is assumed that when the vehicle **20** is travelling behind the vehicle **20b**, the vehicle **20** receives the slip point **120** from the vehicle **20b**. It is assumed that a speed of the vehicle **20** is the same as the vehicle speed of the vehicle **20** in FIG. 1.

(35) The range change unit **220** determines whether the vehicle **20** is travelling on a curved road based on the speed detected by the vehicle speed sensor **26** and the yaw rate detected by the yaw rate sensor **24**. The range change unit **220** sets the fan-shaped detection range **500** with a current position **P5** of the vehicle **20** as a center. Specifically, the detection range **500** is a fan-shaped range with a central angle $\theta 2$ which is surrounded by a straight line $P5P6$, an arc with a radius $L1$, and a straight line $P5P7$. The range change unit **220** sets $\theta 2$ to be larger than $\theta 1$.

(36) As a result, in a case where the vehicle **20** is travelling on a curved road, the detection range in the direction orthogonal to the travelling direction of the vehicle **20** can be set to be wider than a case where the vehicle **20** is travelling on a straight road. As a result, it is possible to avoid missing the detection of the slip point **120**.

(37) FIG. 6 illustrates a data structure of detection range data which is referred to when the range change unit **220** sets a detection range. The detection range data associates a speed and a maximum distance with a central angle in a case where the vehicle is travelling on a straight road and a

central angle in a case where the vehicle is travelling on a curved road.

(38) The speed is a speed of the vehicle **20**. A maximum distance **L1** is a distance from the current position of the vehicle **20** to a farthest position of the detection range. The central angle is an angle formed by left and right straight lines defining the detection range.

(39) As illustrated in FIG. **6**, as the speed is faster, the range change unit **220** sets the maximum distance **L1** to be longer. As the speed is faster, the range change unit **220** sets the central angle to be narrower. In a case where the vehicle **20** is travelling on a curved road, the range change unit **220** sets the central angle to be larger than a case where the vehicle **20** is travelling on a straight road.

(40) As a result, an alert of a slip point with a high risk for the vehicle **20** to travel among slip points detected by other vehicles can be efficiently provided to an occupant of the vehicle **20**. In addition, the alert can be provided when the vehicle speed of the vehicle **20** can be sufficiently slowed down by the time the vehicle **20** reaches the slip point. As a result, it is possible to suppress provision of an unnecessary alert related to the slip point. Therefore, the occupant of the vehicle **20** can avoid feeling cumbersome towards alerts.

(41) In the present embodiment, the slip point information is transmitted and received by the vehicle-to-vehicle communication. However, a mode may be adopted where a server configured to retain the slip point information is further included. For example, when the slip point information transmitted from each vehicle is received, the server may retain the slip point indicated by the slip point information for a predetermined period. The server may transmit information indicating the retained slip point to a vehicle travelling towards the slip point. There may be a case where a vehicle travelling in a position away from another vehicle that has detected the slip point can not receive the slip point information by the vehicle-to-vehicle communication, but with the provision of the server configured to retain the slip point information, the slip point information can be more reliably provided to the vehicle.

(42) Note that the vehicle **20** is a vehicle as an example of transport equipment. The vehicle may be an automobile such as an automobile including an internal combustion engine, an electric vehicle, and a fuel cell vehicle (FCV). The automobile includes a bus, a truck, a two-wheeled vehicle, or the like. The vehicle may be a saddle type vehicle or the like, and may be a motorcycle. The transport equipment includes, in addition to the vehicle, equipment such as an aircraft including an unmanned aerial vehicle, and a ship. The transport equipment may be any equipment for transporting people or items. The transport equipment is an example of the movable object. The movable object is not limited to the transport equipment but may be any movable equipment.

(43) FIG. **7** illustrates an example of a computer **2000** in which a plurality of embodiments of the present invention may be entirely or partially embodied. Programs installed in the computer **2000** can cause the computer **2000** to: function as a system according to the embodiments or each unit of the system or an apparatus such as the assistance apparatus or each unit of the apparatus; execute operations associated with the system or each unit of the system or the apparatus or each unit of the apparatus; and/or execute a process according to the embodiments or steps of the process. Such a program may be executed by a CPU **2012** to cause the computer **2000** to perform certain operations associated with the processing procedures described herein and some of or all of the blocks in the block diagrams.

(44) The computer **2000** according to the present embodiment includes the CPU **2012** and a RAM **2014**, which are mutually connected by a host controller **2010**. The computer **2000** also includes a ROM **2026**, a flash memory **2024**, a communication interface **2022**, and an input/output chip **2040**. The ROM **2026**, the flash memory **2024**, the communication interface **2022**, and the input/output chip **2040** are connected to the host controller **2010** via an input/output controller **2020**.

(45) The CPU **2012** operates according to programs stored in the ROM **2026** and the RAM **2014**, thereby controlling each unit.

(46) The communication interface **2022** communicates with other electronic devices via a network.

The flash memory **2024** stores programs and data used by the CPU **2012** within the computer **2000**. The ROM **2026** stores therein a boot program or the like executed by the computer **2000** at the time of activation, and/or a program depending on the hardware of the computer **2000**. The input/output chip **2040** may connect various input/output units such as a keyboard, a mouse, and a monitor to the input/output controller **2020** via input/output ports such as a serial port, a parallel port, a keyboard port, a mouse port, a monitor port, a USB port, and a HDMI (registered trademark) port. (47) A program is provided via a network or computer readable storage media such as a CD-ROM, a DVD-ROM, or a memory card. The RAM **2014**, the ROM **2026**, or the flash memory **2024** is an example of the computer readable storage medium. Programs are installed in the flash memory **2024**, the RAM **2014**, or the ROM **2026** and executed by the CPU **2012**. The information processing written in these programs is read by the computer **2000**, and thereby cooperation between a program and the above-described various types of hardware resources is achieved. An apparatus or method may be constituted by carrying out the operation or processing of information by using the computer **2000**.

(48) For example, when communication is carried out between the computer **2000** and an external device, the CPU **2012** may execute a communication program loaded onto the RAM **2014** to instruct communication processing to the communication interface **2022**, based on the processing written in the communication program. The communication interface **2022**, under control of the CPU **2012**, reads transmission data stored on transmission buffering regions provided in recording media such as the RAM **2014** and the flash memory **2024**, and transmits the read transmission data to a network and writes reception data received from a network to reception buffering regions or the like provided on the recording media.

(49) In addition, the CPU **2012** may cause all or a necessary portion of a file or a database to be read into the RAM **2014**, the file or the database having been stored in a recording medium such as the flash memory **2024**, etc., and perform various types of processing on the data on the RAM **2014**. The CPU **2012** may then write back the processed data to the recording medium.

(50) Various types of information, such as various types of programs, data, tables, and databases, may be stored in the recording medium to undergo information processing. The CPU **2012** may perform various types of processing on the data read from the RAM **2014**, which includes various types of operations, information processing, conditional judging, conditional branch, unconditional branch, search/replace of information, etc., as described herein and designated by an instruction sequence of programs, and writes the result back to the RAM **2014**. In addition, the CPU **2012** may search for information in a file, a database, etc., in the recording medium. For example, when a plurality of entries, each having an attribute value of a first attribute associated with an attribute value of a second attribute, are stored in the recording medium, the CPU **2012** may search for an entry matching the condition whose attribute value of the first attribute is designated, from among the plurality of entries, and read the attribute value of the second attribute stored in the entry, thereby acquiring the attribute value of the second attribute associated with the first attribute satisfying the predetermined condition.

(51) The program or software module described above may be stored in a computer readable storage medium on the computer **2000** or near the computer **2000**. A recording medium such as a hard disk or a RAM provided in a server system connected to a dedicated communication network or the Internet can be used as the computer readable storage media. A program stored in the computer readable storage medium may be provided to the computer **2000** via a network.

(52) The programs installed onto the computer **2000** for causing the computer **2000** to function as the assistance apparatus **40** may instruct the CPU **2012** or the like to cause the computer **2000** to function as each unit of the assistance apparatus **40**. The information processing written in these programs are read by the computer **2000** to cause the computer to function as each unit of the assistance apparatus **40**, which is specific means realized by the cooperation of software and the various types of hardware resources described above. These specific means implement operations

or processing of information according to the intended use of the computer **2000** in the present embodiment, and the assistance apparatus **40** is thereby constructed to be specific for the intended use.

(53) Various embodiments have been described by referring to the block diagrams and the like. In the block diagram, each block may represent (1) a step of a process in which an operation is executed, or (2) each unit of the apparatus having a role of executing the operation. Certain steps and each unit may be implemented by dedicated circuitry, programmable circuitry supplied with computer readable instructions stored on computer readable storage media, and/or processors supplied with computer readable instructions stored on computer readable storage media. Dedicated circuitry may include digital and/or analog hardware circuits and may include integrated circuits (IC) and/or discrete circuits. Programmable circuitry may include reconfigurable hardware circuits including logical AND, logical OR, logical XOR, logical NAND, logical NOR, and other logical operations, flip-flops, registers, memory elements, etc., such as field-programmable gate arrays (FPGA), programmable logic arrays (PLA), etc.

(54) Computer readable storage media may include any tangible device that can store instructions for execution by a suitable device, such that the computer readable storage medium having instructions stored therein forms at least a portion of an article of manufacture including instructions which can be executed to create means for performing a processing procedure or operations specified in the block diagrams. Examples of the computer readable storage medium may include an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, and the like. More specific examples of the computer readable storage medium may include a floppy disk, a diskette, a hard disk, a random access memory (RAM), a read only memory (ROM), an erasable programmable read only memory (EPROM or flash memory), an electrically erasable programmable read only memory (EEPROM), a static random access memory (SRAM), a compact disk read only memory (CD-ROM), a digital versatile disk (DVD), a Blu-ray (registered trademark) disk, a memory stick, an integrated circuit card, or the like.

(55) The computer readable instruction may include an assembler instruction, an instruction-set-architecture (ISA) instruction, a machine instruction, a machine dependent instruction, a microcode, a firmware instruction, state-setting data, or either of source code or object code written in any combination of one or more programming languages including an object-oriented programming language such as Smalltalk (registered trademark), JAVA (registered trademark), and C++, and a conventional procedural programming language such as a “C” programming language or a similar programming language.

(56) Computer readable instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus, or to programmable circuitry, locally or via a local area network (LAN), wide area network (WAN) such as the Internet, etc., to execute the computer readable instructions to provide means for performing described processing procedure or operations specified in the block diagrams. Examples of the processor include a computer processor, a processing unit, a microprocessor, a digital signal processor, a controller, a microcontroller, and the like.

(57) While the present invention has been described with the embodiments, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations and improvements can be added to the above-described embodiments. It is also apparent from the description of the claims that the embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention.

(58) The operations, procedures, steps, and stages of each process performed by an apparatus, system, program, and method shown in the claims, specification, or drawings can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the

output from a previous process is not used in a later process. Even if the process flow is described using phrases such as “first” or “next” in the claims, specification, or drawings, it does not necessarily mean that the process must be performed in this order.

EXPLANATION OF REFERENCES

(59) **20**: vehicle; **29**: sensor; **24**: yaw rate sensor; **25**: GNSS receiver; **26**: vehicle speed sensor; **40**: assistance apparatus; **200**: processing unit; **210**: calculation unit; **220**: range change unit; **230**: detection unit; **240**: assistance control unit; **250**: communication apparatus; **251**: reception unit; **252**: transmission unit; **270**: alert apparatus; **290**: storage unit; **2000**: computer; **2010**: host controller; **2012**: CPU; **2014**: RAM; **2020**: input/output controller; **2022**: communication interface; **2024**: flash memory; **2026**: ROM; **2040**: input/output chip.

Claims

1. An assistance apparatus included in a vehicle, the assistance apparatus comprising: a reception unit configured to receive information indicating a travelling hazardous point which is transmitted from an outside of the vehicle; a calculation unit configured to calculate, in response to receiving the information indicating the travelling hazardous point by the reception unit, a distance and an orientation from the vehicle to the travelling hazardous point based on the travelling hazardous point and a current position of the vehicle; a detection unit configured to determine whether the travelling hazardous point indicated by the information received by the reception unit is included inside a detection range, wherein the detection range is set, based on the distance and the orientation calculated by the calculation unit, between a first distance from the current position of the vehicle and a second distance from the current position of the vehicle, wherein the second distance is larger than the first distance; and an assistance control unit configured to perform travelling assistance when the detection unit determines that the travelling hazardous point is included inside the detection range, wherein the first distance is a travel distance required for a speed of the vehicle to be slowed down to a predetermined speed or less by a time the vehicle reaches the travelling hazardous point from the current position of the vehicle.
2. The assistance apparatus according to claim 1, comprising: a range change unit configured to change a width of the detection range depending on whether a travel path of the vehicle is a straight road or a curved road.
3. The assistance apparatus according to claim 2, wherein in a case where the travel path of the vehicle is a curved road, the range change unit is configured to set the detection range to be wider than a case where the travel path of the vehicle is a straight road.
4. The assistance apparatus according to claim 3, further comprising: a transmission unit configured to transmit, when it is sensed that a point at which the vehicle has travelled is hazardous for travelling, information indicating the point that is sensed to be hazardous for travelling as information indicating a travelling hazardous point to an outside of the vehicle, wherein the assistance control unit is configured to suppress the travelling assistance when it is sensed that a point at which the vehicle is travelling is hazardous for travelling.
5. The assistance apparatus according to claim 2, wherein in a case where the travel path of the vehicle is a curved road, the range change unit is configured to set a spread angle of the detection range from the current position of the vehicle to be larger than a case where the travel path of the vehicle is a straight road.
6. The assistance apparatus according to claim 5, further comprising: a transmission unit configured to transmit, when it is sensed that a point at which the vehicle has travelled is hazardous for travelling, information indicating the point that is sensed to be hazardous for travelling as information indicating a travelling hazardous point to an outside of the vehicle, wherein the assistance control unit is configured to suppress the travelling assistance when it is sensed that a point at which the vehicle is travelling is hazardous for travelling.

7. The assistance apparatus according to claim 2, further comprising: a transmission unit configured to transmit, when it is sensed that a point at which the vehicle has travelled is hazardous for travelling, information indicating the point that is sensed to be hazardous for travelling as information indicating a travelling hazardous point to an outside of the vehicle, wherein the assistance control unit is configured to suppress the travelling assistance when it is sensed that a point at which the vehicle is travelling is hazardous for travelling.
8. The assistance apparatus according to claim 2, wherein when performing the travelling assistance, even when the reception unit receives information indicating a new travelling hazardous point, the assistance control unit is configured to continue the travelling assistance until the vehicle passes through the travelling hazardous point and to suppress travelling assistance based on the information indicating the new travelling hazardous point.
9. The assistance apparatus according to claim 1, further comprising: a transmission unit configured to transmit, when it is sensed that a point at which the vehicle has travelled is hazardous for travelling, information indicating the point that is sensed to be hazardous for travelling as information indicating a travelling hazardous point to an outside of the vehicle, wherein the assistance control unit is configured to suppress the travelling assistance when it is sensed that a point at which the vehicle is travelling is hazardous for travelling.
10. The assistance apparatus according to claim 1, wherein when performing the travelling assistance, even when the reception unit receives information indicating a new travelling hazardous point, the assistance control unit is configured to continue the travelling assistance until the vehicle passes through the travelling hazardous point and to suppress travelling assistance based on the information indicating the new travelling hazardous point.
11. The assistance apparatus according to claim 1, wherein the reception unit is configured to receive the information indicating the travelling hazardous point transmitted from one or more other vehicles via a vehicle-to-vehicle communication.
12. An assistance apparatus comprising: a reception unit configured to receive information indicating a travelling hazardous point which is transmitted from an outside of a vehicle; a calculation unit configured to calculate a distance and an orientation from the vehicle to the travelling hazardous point based on the travelling hazardous point and a current position of the vehicle; a detection unit configured to detect the travelling hazardous point inside a detection range set within a range in which a distance from the vehicle is a first distance or more and a second distance or less; an assistance control unit configured to perform travelling assistance when the travelling hazardous point is detected inside the detection range; and a range change unit configured to set a spread angle of the detection range from the current position of the vehicle to be larger and set the second distance to be longer as a travelling speed of the vehicle is faster, wherein the first distance is a travel distance required for a speed of the vehicle to be slowed down to a predetermined speed or less by a time the vehicle reaches the travelling hazardous point from the current position of the vehicle.
13. The assistance apparatus according to claim 12, comprising: a range change unit configured to change a width of the detection range depending on whether a travel path of the vehicle is a straight road or a curved road.
14. The assistance apparatus according to claim 13, wherein in a case where the travel path of the vehicle is a curved road, the range change unit is configured to set the detection range to be wider than a case where the travel path of the vehicle is a straight road.
15. The assistance apparatus according to claim 14, wherein in a case where the travel path of the vehicle is a curved road, the range change unit is configured to set a spread angle of the detection range from the current position of the vehicle to be larger than a case where the travel path of the vehicle is a straight road.
16. The assistance apparatus according to claim 13, wherein in a case where the travel path of the vehicle is a curved road, the range change unit is configured to set a spread angle of the detection

range from the current position of the vehicle to be larger than a case where the travel path of the vehicle is a straight road.

17. The assistance apparatus according to claim 12, further comprising: a transmission unit configured to transmit, when it is sensed that a point at which the vehicle has travelled is hazardous for travelling, information indicating the point that is sensed to be hazardous for travelling as information indicating a travelling hazardous point to an outside of the vehicle, wherein the assistance control unit is configured to suppress the travelling assistance when it is sensed that a point at which the vehicle is travelling is hazardous for travelling.

18. The assistance apparatus according to claim 12, wherein when performing the travelling assistance, even when the reception unit receives information indicating a new travelling hazardous point, the assistance control unit is configured to continue the travelling assistance until the vehicle passes through the travelling hazardous point and to suppress travelling assistance based on the information indicating the new travelling hazardous point.

19. A vehicle comprising an assistance apparatus including: a reception unit configured to receive information indicating a travelling hazardous point which is transmitted from an outside of the vehicle; a calculation unit configured to calculate, in response to receiving the information indicating the travelling hazardous point by the reception unit, a distance and an orientation from the vehicle to the travelling hazardous point based on the travelling hazardous point and a current position of the vehicle; a detection unit configured to determine whether the travelling hazardous point indicated by the information received by the reception unit is included inside a detection range, wherein the detection range is set, based on the distance and the orientation calculated by the calculation unit, between a first distance from the current position of the vehicle and a second distance from the current position of the vehicle, wherein the second distance is larger than the first distance; and an assistance control unit configured to perform travelling assistance when the detection unit determines that the travelling hazardous point is included inside the detection range, wherein the first distance is a travel distance required for a speed of the vehicle to be slowed down to a predetermined speed or less by a time the vehicle reaches the travelling hazardous point from the current position of the vehicle.

20. A non-transitory computer readable storage medium having stored thereon a program, the program causing a computer to function as: a reception unit configured to receive information indicating a travelling hazardous point which is transmitted from an outside of a vehicle; a calculation unit configured to calculate, in response to receiving the information indicating the travelling hazardous point by the reception unit, a distance and an orientation from the vehicle to the travelling hazardous point based on the travelling hazardous point and a current position of the vehicle; a detection unit configured to determine whether the travelling hazardous point indicated by the information received by the reception unit is included inside a detection range, wherein the detection range is set, based on the distance and the orientation calculated by the calculation unit, between a first distance from the current position of the vehicle and a second distance from the current position of the vehicle, wherein the second distance is larger than the first distance; and an assistance control unit configured to perform travelling assistance when the detection unit determines that the travelling hazardous point is included inside the detection range, wherein the first distance is a travel distance required for a speed of the vehicle to be slowed down to a predetermined speed or less by a time the vehicle reaches the travelling hazardous point from the current position of the vehicle.

21. An assistance method comprising: receiving information indicating a travelling hazardous point which is transmitted from an outside of a vehicle; calculating, in response to receiving the information indicating the travelling hazardous point, a distance and an orientation from the vehicle to the travelling hazardous point based on the travelling hazardous point and a current position of the vehicle; determining whether the travelling hazardous point indicated by the received information is included inside a detection range, wherein the detection range is set, based on the

calculated distance and the calculated orientation, between a first distance from the current position of the vehicle and a second distance from the current position of the vehicle, wherein the second distance is larger than the first distance; and performing travelling assistance when the travelling hazardous point is determined to be included inside the detection range, wherein the first distance is a travel distance required for a speed of the vehicle to be slowed down to a predetermined speed or less by a time the vehicle reaches the travelling hazardous point from the current position of the vehicle.
